CAFs-derived exosomes inhibits ferroptosis via GALNT14-mediated O-GalNAcylation of SLC7A11 in colorectal cencer

huayang pan

panhuayang88@hrbmu.edu.cn

First Affiliated Hospital of Harbin Medical University https://orcid.org/0009-0003-2549-2734

Jialu Guan

Lingfeng Guo

Shuo Ning

Wenxi Zhao

Tong Wang

Jingjng Wang

Zhen Qi

Jingjing Huang

Jianqi Wang

Yaunfeng Gong

Tiemin Pei

Qinghui Meng

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Abstract

In the tumor microenvironment (TME), cancer-associated fibroblasts (CAFs), the dominant stromal component, actively shape cancer progression through exosomal communication. Here, we identify *hsa_circ_0003892* (*circLDLR*), a CAF-derived circRNA, as a key factor linked to poor prognosis in colorectal cancer (CRC). During CAF−CRC interaction, *circLDLR* is packaged into exosomes and transferred to tumor cells, enhancing proliferation and metastasis largely by reducing ferroptosis susceptibility. Mechanistic exploration revealed that *circLDLR* stabilizes Polypeptide N-Acetylgalactosaminyltransferase 14 (GALNT14) by shielding it from ZNRF2-mediated ubiquitin degradation, leading to elevated GALNT14 protein levels. Elevated GALNT14 promotes O-GalNAcylation of Solute Carrier Family 7 Member 11 (SLC7A11) at Ser26, facilitating its membrane localization, thereby suppressing ferroptosis in CRC cells. Moreover, EIF4A3, an RNA-binding protein (RBP), contributes to *circLDLR* biogenesis within CAFs. Taken together, our study reveals that CAFs-derived *circLDLR* can confer ferroptosis resistance and boost the progression of CRC, which are mainly dependent on increasing the stability of GALNT14 and enhancing O-GalNAcylation-mediated membrane localization of SLC7A11, thus, disrupting circLDLR transfer between CAFs and CRC cells may offer a promising approach for CRC therapy.

INTRODUCTION

CRC remains one of the most prevalent and lethal malignancies worldwide, with its progression and metastasis being major contributors to patient mortality[1]. Components of the tumor microenvironment (TME) interact with malignant cells, contributing to the hallmark of cancer[2]. CAFs are among the most abundant and versatile components within the TME. They are histologically prominent and biologically important in CRC initiation, progression, and metastasis[3, 4]. Growing evidence have demonstrated that CAFs can promote cancer cells proliferation, invasion, and resistance to therapy via secreting exosomes that facilitate intercellular communication within the TME[5]. Exosomes are critical cellular communicators. Accumulating evidences have reported that exosomes secreted by stromal cells play a pivotal role in cancer progression by encapsulating a variety of proteins, lipid, mRNAs, microRNAs, lncRNAs, circRNA and transferring to cancer cells[6, 7]. Among the molecules carried by exosomes, circRNAs have attracted significant attention due to their characteristics.

Beyond the role as miRNA sponges, circRNAs have been shown to regulate protein stability, modulate signaling pathways, and influence cellular processes such as proliferation, migration, and apoptosis[8, 9]. However, the mechanism that how circRNAs contribute to CRC progression, particularly through interactions with the TME, warrants further investigation.

Ferroptosis is an iron-dependent programmed death pathways, its molecular properties distinguish it from other types of programmed cell death[10]. Dysregulation of ferroptosis has been implicated in the progression of various diseases, including cancer[11].

Glycosylation represents a major form of protein post-translational modification (PTM), in which polysaccharides are transferred to specific amino acid residues in proteins via glycosyltransferases. Glycosylation plays crucial roles in protein stabilization. O-glycosylation is polysaccharides bonded to the hydroxyl group of the oxygen atom of serine or threonine residues. O-GalNAcylation, also known as O-glycosylation mucins, is initiated by polypeptide N-acetylgalactosaminyl-transferase. This complex mechanism involves more than 20 different peptide GalNAc transferase enzymes. O-GalNAcylation is expressed across multiple tumor types and plays critical roles in tumor metastasis, immune evasion and metabolic reprogramming.

In this study, we identified *circLDLR* as a significantly upregulated circRNA driving the progression of CRC. We further demonstrated that *circLDLR* is highly expressed in CAFs, where it promotes CRC progression through exosomes-packaged *circLDLR* is internalized by CRC cells. Mechanistically, *circLDLR* stabilizes GALNT14, a key regulator of protein O-GalNAcylation, by inhibiting its ubiquitin-mediated degradation, thereby enhancing the membrane localization of SLC7A11 and suppressing ferroptosis in CRC cells.

RESULTS

The expression and characteristics of circLDLR

To identify abnormally expressed circRNAs in CRC, two independent CRC datasets (GSE126094, GSE147597) from the Gene Expression Omnibus (GEO) database were selected and analyzed. 7 differentially expressed circRNAs upregulated in CRC were found (Fig. 1A, Fig. S1). Subsequently, thoese circRNAs were chosen for further detection. Using qRT-PCR analysis of 43 pairs of CRC samples, it was validated that *hsa_circ_0003892* exhibited the most significant differential expression (Fig. 1B). To further analyze the relationship between *circLDLR* expression and clinicopathologic features in CRC patients, we examined 43 CRC samples and found that high *circLDLR* expression was positively correlated with tumor size, lymph node metastasis, and stage (Fig. S2A, Table S1). According to circBase, *circLDLR* is located on human chromosome 19 and is generated from the back-splicing of 13–16 exons of the LDLR gene. The back-splice junction site was confirmed by Sanger sequencing in CRC cells (Fig. 1C). Then, we designed divergent primers and convergent primers, which were used to amplify in gDNA and cDNA, respectively. As shown in Fig. 1D, agarose gel electrophoresis revealed that *circLDLR* could only be amplified from cDNA using the divergent primers. Additionally, due to the closed loop structure of *circLDLR*, its expression level was not affected by RNase R treatment, whereas the expression of linear LDLR significantly decreased (Fig. 1E).

To better understand the source of *circLDLR*, we performed FISH analysis on CRC tissues and adjacent normal tissues, finding that *circLDLR* was predominantly localized in the cytoplasm of CRC stromal cells, in sharp contrast to the epithelium (Fig. 1F). *CircLDLR* FISH and Vimentin immunofluorescence co-localization experiments conclusively demonstrated that *circLDLR* is predominantly localized within CAFs of tumor tissues (Fig. 1G). CAFs and NFs were isolated from CRC tissues and adjacent normal tissues, and adherent in culture with aspindle-shaped morphology (Fig. 1H). Herein, Western blot was performed to further confirm the purity and phenotype of NFs and CAFs. The results showed that in CAFs, the expression levels of FAP and α-SMA were notably higher than in NFs, while no differential expression of Vimentin (Fig. 1I, K). Compared with other cells, FISH revealed that *circLDLR* had higher expression level in CAFs (Fig. 1J). Next, CAFs and NFs were isolated from 43 pairs of CRC patients, and *circLDLR* were detected by qRT-PCR. It was found that the expression level of *circLDLR* in CAFs was significantly increased compared to NFs (Fig. S2B). Therefore, elevated *circLDLR* in CAFs may be significantly related to the occurrence and development of CRC.

The aberrant expression of circLDLR in CAFs exacerbates CRC progression

The level of *circLDLR* in primary fibroblasts and CRC cells lines were detected by qRT-PCR and showed that the level was highest in CAFs. Among the CRC cell lines, HCT116 exhibited relatively high expression, while RKO show relatively low expression (Fig. S3A). To further identify the function of CAFs in CRC, CRC cells were co-cultured with CAFs and NFs. It was found that CAFs dramatically promoted the proliferation, migration and invasion abilities of CRC cells. Furthermore, compared with the NFs group, CAFs co-culture notably potentiated the viability of CRC cells (Fig. S3B-D).

To further verify whether the oncogenic effect of CAFs was mainly relied on *circLDLR*. Firstly, the *circLDLR* in CAFs was deleted or overexpressed, and then transfected CAFs were co-cultured with CRC cells. As expected, overexpression of *circLDLR* in CAFs notably facilitated the growth of CRC cells, while *circLDLR* deletion generated the opposite effect (Fig. S3E-G). Next, qRT-PCR was utilized to determine *circLDLR* in CAFs and co-cultured CRC cells. As shown in Fig. S3H, I, silencing *circLDLR* in CAFs notably reduced the expression of *circLDLR* in CAFs and co-cultured CRC cells, whereas ectopic expression of *circLDLR* in CAFs exhibited contrary effects. Taken together, we hypothesized that *circLDLR* may shuttle from CAFs to CRC cells, thereby aggravated the malignant behaviors of CRC cells.

Exosomal circLDLR shuttles from CAFs to CRC cells

Given that recent evidence have reported CAFs could exacerbate cancer progression[12], exosomes may play an important role in crosstalk between CAFs and cancer cells[13], which inspired us to suppose whether exosomes are essential in *circLDLR* transfer. Exosomes in CAFs/NFs conditioned medium were purified by ultracentrifugation then confirmed by transmission electron microscopy and nanoparticle tracking analysis (NTA) (Fig. 2A). The exosomes were observed to be cup-shaped structure with diameters of 100 nm. Further, exosome markers CD81, CD63, TSG101 and Calnexin were detected by Western blotting analysis (Fig. 2B). CAFs derived exosomes were labeled with PKH67. Subsequently, PKH67-labeled exosomes were internalized by co-cultured CRC cells. (Fig. 2C). Further, CRC cells were co-cultured with CAFs transfected with cy3-tagged *circLDLR*. As shown in Fig. 2D, E, the fluorescently labeled *circLDLR* was notably increased in CRC cells, but the elevation could be abrogated by GW4869 (an inhibitor of exosomes secretion). The level of *circLDLR* in GW4869-treated CAFs remained stable (Fig. 2F). Additionally, in order to better characterize the biological function of *circLDLR* in CRC cells, *circLDLR* of CAFs was overexpressed or deleted, then CAFs derived exosomes were purified and incubated with CRC cells, as we expected, in ablation/overexpression group, the promoting effect on the progression of CRC cells correspondingly attenuated or potentiated (Fig. 2G, H, Fig. S4A, B), consistent with the results of CAFs co-cultured CRC cells. Further, the results of qRT-PCR revealed that overexpression of *circLDLR* increased levels of *circLDLR* in CAF exosomes and cancer cells, whereas deletion of *circLDLR* exerted the opposite effect (Fig. 2I, J). Taken together, these results demonstrates that CAFs-derived exosomal *circLDLR* can be transferred to CRC cells and facilitate progression of CRC cells.

CAFs-derived circLDLR facilitates the progression of CRC cells by suppressing ferroptosis

To better characterize of CAFs-derived *circLDLR* in CRC, *circLDLR* were deleted in CAFs, and then exosomes enriched from CAFs were co-cultured with CRC cells. Subsequently, those cells were treated with various programmed cell death inhibitors (including ferroptosis, necroptosis, apoptosis and autophagy) to explore the possible cell death pattern that modulated by *circLDLR*. Intriguingly, Lip-1 and Fer-1, two inhibitors of ferroptosis, reversed the increase death of CRC cells when added CAFs (sh-*circLDLR*)-derived exosomes, but not Nec-1(necroptosis inhibitor), Z-VAD(apoptosis inhibitor) and CQ (autophagy inhibitor) (Fig. 3A). These results

indicated that *circLDLR* of CAFs-derived exosomes might be involved in the progresssion of ferroptosis in CRC cells. Furthermore, CRC cells were treated with erastin and detected the proliferation ability, the result showed that CAFs (sh-NC)-derived exosomes and Lip-1 significantly decreased erastin-induced cell death, but not CAFs (sh-*circLDLR*)-derived exosomes (Fig. 3B). Additionally, we checked the concentration of the relative intracellular lipid peroxidation, mitochondrial superoxide (MitoSOX) and Mitochondrial Membrane Potential (MMP), CAFs (sh-NC)-derived exosomes and Lip-1 increased those levels of HCT116 and RKO cells (Fig. 3C-E, Fig. S5A-C). As we all known, the intracellular Fe^{2+} and Iron are a pool of redox-active iron and essentional for triggering oxidative damage. As shown in Fig. 3F, G and Fig. S5D, E, Fe^{2+} and Iron levels were higher in erastin-treated cells than control, whereas CAFs (sh-NC)-derived exosomes and Lip-1 could reverse these alterations. Transmission electron microscope (TEM) analysis further verified that CAFs (sh-NC)-derived exosomes particularly restored typical morphological features of ferroptosis, including shrunken mitochondria with elevated membrane density (Fig. 3H, Fig. S5F). Invasion, migration and colony formation assays also validated the effect of CAFs (sh-NC)-derived exosomes (Fig. 3I, J Fig. S5G, H). Taken together, we conclude that the *circLDLR* from CAFs-derived exosomes promote CRC cells growth via conferring ferroptosis resistance.

CircLDLR stabilizes GALNT14 via protecting it against ZNRF2-mediated ubiquitin/proteasome-mediated degradation

To dissect how *circLDLR* regulates cell ferroptosis, we first determined the subcellular localization of *circLDLR* by nuclear and cytoplasmic fractionation as well as FISH examination. The results indicated that *circLDLR* predominantly localized in the cytoplasm of CRC cells (Fig. 4A). The results of RIP assays showed that the circular RNA ciRS-7, function as competitive endogenous RNA (ceRNA), was significantly enriched by AGO2, but not *circLDLR* (Fig. 4B). Then, we conducted circRNA pull-down and MS assays to examine potential proteins bind with *circLDLR* (Fig. 4C, Table S2). Among those proteins, GALNT14 aroused our interest, and the binding between *circLDLR* and GALNT14 was further confirmed by RIP assays (Fig. 4D). Then, we performed IHC staining to reveal that GALNT14 presented the higher level in CRC tissue and predominantly localized in the cytoplasm (Fig. 4E). In addition, we performed RNA FISH-immunofluorescence analysis and found *circLDLR* co-localized with GALNT14 in the cytoplasm (Fig. 4F). When *circLDLR* was deleted/overexpressed in CRC cells, as expected, the expression of *circLDLR* was positiveiy correlated with GALNT14 protein level but not mRNA level (Fig. 4G). To further address the molecular basis underlying the interaction between *circLDLR* and GALNT14, we designed four mutants based on the structural composition of *circLDLR* (formed by backsplicing of four exons). Mutation of the *circLDLR* 296-466nt significantly decreased GALNT14 enrichment by *circLDLR* (Fig. 4H), indicating that the 296-466nt sequences is essential for the *circLDLR*-GALNT14 interaction. Next, we investigated the mechanism of *circLDLR* regulating GALNT14 level. Firstiy, CRC cells were treated with CHX to block protein synthesis for indicated times. The result of western blot revealed that *circLDLR* could stabilize GALNT14. (Fig. 4I). Next, we found that the effect of circLDLR depletion on GALNT14 protein levels was not affected by the lysosome inhibitor CQ but could be abolished by proteasome inhibitors MG-132 (Fig. 4J). Additionally, overexpression of wild-type circLDLR decreased the level of Ubiquitination on GALNT14, whereas overexpression of the mutant circLDLR had no effect (Fig. 4K). Thus, these findings suggest that *circLDLR* plays a central role in GALNT14 protein degradation through the ubiquitination/proteasome pathway. K48-, K29-and K63-linked polyubiquitin chains are three main types of polyubiquitin linkage. We constructed vectors carrying K48-only, K29-only and K63-only respectively and bearing HA tags. Ubiquitination assays suggested that K48-linked polyubiquitin of GALNT14 was reduced by *circLDLR* overexpression, while other polyubiquitin linkage had no evidently alteration (Fig. 4L). In sum, these findings validate that *circLDLR* promotes GALNT14 expression via protecting it from degradation of the K48-linked ubiquitin-proteasome pathway.

Considering the E3 ubiquitin ligase is essential for the ubiquitination reaction, UbiBrowser 2.0 was applied to determine the potential E3 ligases might be involved in GALNT14 degradation (Fig. S6A). Meanwhile, STARBASE database showed that ZNRF2 have a lower level in cancer than in normal tissue (Fig. S6B). Thus, we supposed that ZNRF2 is the E3 ubiquitin ligase for GALNT14. To confirm whether ZNRF2 participates in regulating the ubiquitination effect of GALNT14, CO-IP assays were conducted, and the reciprocal interaction between ZNRF2 and GALNT14 was confrmed (Fig. 4M). The results of western blot showed that ZNRF2 overexpression significantly reduced the level of GALNT14, and could be rescued by MG-132 (Fig. 4N). CRC cells were treated with CHX for indicated times, and western blot analysis was implemented. The result showed that ZNRF2 overexpression also shortened the half-life of GALNT14 protein (Fig. 4O). In addition, we transfected K63-only, K48-only, K63R, K48R mutant ubiquitin and overexpressed ZNRF2, and found that ZNRF2 only dictated the K48-linked polyubiquitin of GALNT14 (Fig. 4P). Therefore, these results suggest that ZNRF2 is the E3 ubiquitin ligase which binds with GALNT14, and degrades them through the K48-linked polyubiquitin of GALNT14.

CircLDLR confers ferroptosis resistance by upregulating GALNT14

According to the results in Fig. 4G and 4P, *circLDLR* and ZNRF2 both affect the ubiquitination degradation of GALNT14 through K48-linked ubiquitination. So, to further verity whether the effect of ZNRF2 on GALNT14 protein could be influenced by *circLDLR*, we conducted CO-IP and PLA assays. As we expected, the interaction between ZNRF2 and GALNT14 was disrupted by *circLDLR* overexpression (Fig. 5A, B). *CircLDLR* ablation could enhance the K48-linked ubiquitination of GALNT14 protein, ZNRF2 silencing could rescue the degradation GALNT14 protein caused by *circLDLR* ablation through the K48-linked ubiquitin–proteasome pathway (Fig. 5C). To further explore whether *circLDLR* influenced cancer cells ferroptosis through GALNT14, CRC cells were treated with erastin. The obtained data showed that migration, invasion and colony abilities were accelerated by *circLDLR*, while this influence was abolished by GALNT14 deletion (Fig. S7A-D). In parallel, several main characteristics of ferroptosis (Lipid peroxidation level, relative levels of MMP and MitoSOX, relative Fe^{2+} and Iron levels) were changed by circLDLR overexpression, and the inhibitory effect of *circLDLR* on ferroptosis could be reversed by GALNT14 detection (Fig. 5D-M). Above all, these evidences concomitantly indicate that *circLDLR* markedly dampens the sensitivity of CRC cells to ferroptosis, which is mainly dependent on GALNT14.

GALNT14 impairs ferroptosis via O-GalNAcylation of SLC7A11 and its membrane localization

To explore the molecular mechanism underlying the inhibition of ferroptosis in CRC cells by GALNT14. Firstly, we overexpressed/deleted GALNT14 in CRC cells, the level of O-GalNAcylation increased/decreased respectively (Fig. S8A, B). Then, GALNT14 overexpression stable transfection CRC cell lines were established,and then co-cultured with ferroptosis inducers of various pathways (including System Xc−, GCL, GPX4, and FSP1). GALNT14 overexpression was found to rescue the ferroptosis only by inducer targeting System Xc−, but not by inducers targeting GCL, GPX4, and FSP1 (Fig. 6A). As we all known, the uptake of cystine and the synthesis of GSH play important roles in the process of ferroptosis[14–16]. The two critical functions of System Xc – were diminished significantly by GALNT14 ablation. When cells were supplemented with the downstream products of System Xc−, Acetylcysteine (NAC) and GSH, ferroptosis induced by GALNT14 ablation was markedly alleviated in CRC cells (Fig. 6B, C). SLC7A11, as the most important functional subunit of System Xc−, proper membrane localization is essential for the functionality of SLC7A11. Previous studies have found that O-GlcNAcylation of SLC7A11 can alter its activity, but the specific mechanism remains unclear. As a member of polypeptide N-acetyl- α-galactosaminyltransferases family, GALNT14 mediated O-GalNAcylation. These findings inspired us to doubt whether SLC7A11 can be regulated by GALNT14.Western blot assay and immunofluorescence assay showed that GALNT14 did not affect the total level of SLC7A11 protein, but altered the membrane localization of SLC7A11 (Fig. 6D, E). CO-IP assays and PLA were conducted in CRC cell and 293T cell, the results confirmed that SLC7A11 could interacte with GALNT14 (Fig. 6F, G). Meanwhile, GALNT14 overexpression/deletion could significantly increase/decrease the O-GalNAcylation level of SLC7A11 (Fig. 6H). These results consistently confirmed the regulatory role of

GALNT14 in the membrane localization of SLC7A11. Further, the mucin-type O-glycosylation sites of SLC7A11 were predicted using NetOGlyc-4.0. The database showed that S8, T9, S26 and T218 might be the O-GalNAcylation sites of SLC7A11, meanwhile, N314, predicted using NetNGlyc-1.0, might be the N-glycosylation site (Fig. 6I, Table S3). Then, S8R, T9R, S26R, T218R mutation vectors were constructed respectively and generating HA-tags, N314R mutation vectors serve as conrol. CO-IP and VVL lectin blot experiments verified that only S26R reduced the interaction of SLC7A11and GALNT14 (Fig. 6J). Then, a serious of assays with these mutant O-GalNAcylation sites of SLC7A11 in GALNT14-OE and SLC7A11-KO stably transfected cells were performed. As we expected, the proliferation, migration and invasion experiments showed that only S26 mutant site could not reverse the promoting effect of ferroptosis initiated by SLC7A11-KO (Fig. S9A, B). CRC cells transfected with S26R also could not rescue the O-GalNAcylation level of SLC7A11 and the level of GSH and Cystine uptake which reduced by SLC7A11-KD (Fig. 6K-M). Western blot and immunofluorescence assays showed that S26 was the key site on the membrane localization of SLC7A11 (Fig. 6N, O). Collectively, the results illustrate that GALNT14 induces O-GalNAcylation of SLC7A11 at the S26 site to promote its membrane localization, and further confers ferroptosis resistance in CRC cells.

Biogenesis of circLDLR is Facilitated by EIF4A3

Prior research had demonstrated that the biogenesis of exon-derived circRNAs can be regulated by RBPs through their interactions with flanking intron sequences of circRNAs[17]. EIF4A3, QKI and FUS had been shown to facilitate circRNA biogenesis[18–20]. To investigate its potential role in *circLDLR* formation, we employed CircInteractome to predict RBP-binding sites within the flanking regions of *circLDLR*. We found 7 putative binding sites for EIF4A3 in the downstream of the *circLDLR* pre-mRNA (Fig. S10A). Due to sequence overlaps, these sites were named as *a-c*, and the sequence on *circLDLR* named as *d* (Fig. S10B). This binding specificity was further validated by RNA pull-down assays (Fig. S10C). Notably, EIF4A3 overexpression significantly upregulated *circLDLR* expression in CAFs, while its deletion reduced *circLDLR* levels (Fig. S10D). To further explore the expression of EIF4A3 in CRC patients, an IHC assay was conducted CRC tissues and adjacent normal tissues, the result revealed that the expression of EIF4A3 was higher in CRC tissues than in adjacent normal tissues (Fig. S10E) and that the expression of EIF4A3 and *circLDLR* was positively correlated in CRC patients (Fig. S10F).Collectively, these results establish that EIF4A3 as a positive regulator of *circLDLR* biogenesis through direct binding to its flanking intronic regions.

CircLDLR in CAFs promotes cancer growth and metastasis in vivo

To further investigate the impact of CAF-exosomes *circLDLR* on cancer progression *in vivo*, two human CRC cell lines were employed to establish tumor model. In subcutaneous xenograft model, *circLDLR* stably overexpressed and deleted CAFs were constructed. Then, CRC cells were co-injected with or without CAFs into the nude mice subcutaneously. As shown in Fig. 7A, B, co-transplanted with CAFs accelerated subcutaneous tumor growth. This pro-tumorigenic effect was enhanced when co-transplanted with CAFs OE-*circLDLR*, whereas co-transplanted with CAFs KD-*circLDLR* displayed the opposite effect. We further investigated the effect of *circLDLR* on CRC metastasis *in vivo*. CAFs OE-*circLDLR* further accelerated the liver metastasis of CRC cells *in vivo*, as indicated by more tumor nodules, and more serious colorectal tissue damage (Fig. 7C). However, CAFs KD-*circLDLR* reversed the lung metastasis of CRC cells *in vivo* (Fig. 7D). In addition, we performed IHC and HE assays. IHC staining showed the result that CAFs sh-circ/CAFs OE-circ decreased/increased the expression of Ki67 and GALNT14 in subcutaneous tumors, whereas the 4HNE exhibited the opposite level (Fig. 7E-G). Moreover, HE staining confirmed the morphology of tumor tissues in lung and liver metastasis groups (Fig. 7H). Collectively, these results revealed that *circLDLR* is essential for CAFs to potentiate the growth and metastasis of CRC *in vivo*.

DISCUSSION

As key cellular components of TME, CAFs play a pivotal role in the TME. Accumulating studies have revealed that CAFs can regulate tumor growth and metastasis through the secretion of non-coding RNAs[21, 22]. Ferroptosis, as an iron-dependent form of regulated cell death, plays a critical role in colorectal cancer progression. Emerging evidence suggests that CAFs are involved in tumor progression, but the precise mechanisms in regulating the interplay between CAFs and ferroptosis remain elusive. In this study, we demonstrate that EIF4A3-facilitated *circLDLR* is predominantly shuttled from CAFs to CRCcells via exosomes, subsequently, *circLDLR* protects against ferroptosis by impairing SLC7A11 membrane localization in CRC cells, thereby facilitating tumor progression and metastasis.

With the advancs of research, their functional roles of circRNA in cellular processes have been increasingly elucidated, particularly their critical function of tumor progression and oncogenesis[8, 9, 23]. A substantial body of research has established that CAFs-derived exosomes influence the progression of various cancers[6, 13, 24]. Notably, exosome-packaged circRNAs contribute significantly to these regulatory effects[7, 25–27]. Beyond functioning as ceRNAs, increasing lines of evidence have demonstrated that circular RNAs can directly bind to proteins and regulate the expression of their target proteins[28–31]. Here, GALNT14 was identified as a promising target of *circLDLR*. We further demonstrated that *circLDLR* binds GALNT14 through its specific functional domain, thereby disrupting the interaction between GALNT14 and the E3 ubiquitin ligase ZNRF2, decelerating ZNRF2-mediated K48-linked polyubiquitination and degradation, ultimately leading to elevated GALNT14 expression. Numerous evidences highlight RBPs as critical regulators of circRNA biogenesis. As a key RBP, Eukaryotic Translation Initiation Factor 4A3 (EIF4A3) has been shown to modulate multiple circRNAs[19, 32, 33]. In this study, bioinformatic predictions identified *circLDLR* as a potential EIF4A3 target. In CAFs, we validated EIF4A3 positive regulation of *circLDLR*. However, whether EIF4A3 directly modulates *circLDLR* back-splicing remains to be further elucidated.

O-linked glycosylation is one common type of Glycosylation[34]. Aberrant glycosylation endows tumor cells with resistance to cell death. Whereas most studies have focused on the role of O-GlcNAcylation in tumors, the function of O-GalNAcylation (mucin-type O-glycosylation) in tumor progression remains largely ignored. Although the two modifications belong to O-linked glycosylation, certain functional differences exist between the two modifications, both modifications target serine or threonine residues. GALNT14 is a pivotal O-glycosyltransferase implicated in tumor progression through its regulation of protein O-glycosylation in various cancers[35, 36]. In HCC, GALNT14-mediated O-GalNAcylation at serine-161 of PHB2 activates the IGF1R cascade, thereby enhancing cell proliferation, migration, and therapy resistance[37]. Similarly, in LUAD, GALNT14-driven O-GalNAcylation suppresses endogenous ROS production to promote tumor aggressiveness[38]. Previous studies have confirmed that ferroptosis, play critical roles in the initiation and progression of colorectal cancer[39–41]. In this study, we identified SLC7A11 as a novel O-GalNAcylated substrate of GALNT14. While prior research primarily focused on expression levels of SLC7A11 (unlike FSP1 studies that emphasized N-myristoylation-dependent membrane localization for ferroptosis regulation[42]), emerging evidence now highlights the critical role of SLC7A11 membrane trafficking in ferroptosis regulation[43, 44]. Recent report indicate that O-GlcNAcylation sustains SLC7A11 activity to suppress ferroptosis[45], even if the mechanistic basis remains unclear. Intriguingly, our work reveals that GALNT14 mediates O-GalNAcylation at S26 of SLC7A11, promoting its membrane localization and functional activation without altering total SLC7A11 abundance—thereby inhibiting ferroptosis and driving CRC progression. This discovery diverges from the canonical paradigm of glycosylation-mediated protein stabilization in ferroptosis regulation, instead establishing a novel mechanism where O-GalNAc modification orchestrates SLC7A11 spatial redistribution to modulate ferroptosis susceptibility. Our finding expands the understanding of O-GalNAcylation in ferroptosis regulation and provide new directions for targeting SLC7A11 membrane dynamics in cancer therapy.

In conclusion, our study reveals a novel CAF-to-CRC signaling axis: EIF4A3 upregulates *circLDLR* in CAFs, where *circLDLR* is packaged into exosomes and transferred to CRC cells. Within CRC cells, *circLDLR* binds and stabilizes GALNT14 by protecting against ZNRF2-mediated ubiquitination and degradation. Then, the accumulated GALNT14 catalyzes O-GalNAcylation at S26 site of SLC7A11, promoting its membrane localization to inhibit ferroptosis—ultimately driving CRC growth and metastasis. These findings highlight the importance of in tumor-stromal interactions and provide a rational basis for developing targeted therapies to disrupt the CAFs-CRC cells crosstalk.

METHODS

Cell culture

Human HCT116, HT29, RKO, LOVO, SW620, SW480 and HEK 293T cells were purchased from Procell (Wuhan,China). Human NCM460 cell was purchased from Fuheng Biology (Shanghai, China). Human HCT116 and HT29 cell lines were cultured in McCoy's 5A Complete medium (Procell, #PM150710B). Human RKO cell line was cultured in MEM medium (Gibco,#11095080). Human LOVO cell line were cultured in Ham's F-12K complete medium (Procell, #CM-0144). SW620, SW480, HEK 293T and NCM460 cells were cultured in DMEM medium (Gibco,#11965092). MEM and DMEM medium plus 10% fetal bovine serum (FBS) (TIANHANG, #13011 – 8611) and 1% penicillin/streptomycin (SEVEN BIOTECH, #SC118-01). Cell lines were incubated at 37°C with 5% CO_2. Mycoplasma contamination was confirmed to be negative in cells. After thawing the frozen stocks, cells were generally passaged fewer than 10 times.

Human CRC tissue specimens

Primary CRC specimens and paired non-cancerous tissues were obtained from the First Affiliated Hospital of Harbin Medical University. The characteristics of the patients are listed in Table S1. The study was approved by the Ethics Committee of the First Affiliated Hospital of Harbin Medical University, and written informed consent was provided by the patients.

Isolation and culture of patient-derived NFs and CAFs

CAFs were isolated from CRC tissues of patients, and NFs were separated from the paired adjacent normal area. Briefly, the fresh paired tissues were washed three times with PBS containing 100 U/ ml penicillin and 100 μg/ml streptomycin. Then, the tissues were chopped into small pieces and digested in collagenase II (Gibco, #17101015) at 37°C. All cells were collected and resuspended in DMEM/F12 medium (Gibco, #11320033) containing 15% FBS. The suspensions were fostered in DMEM/F12 medium supplemented with 15% FBS for about 48 hours. After stable adherence of the cells, DMEM/F12 medium was used to wash out the excess tissues and dead cells, then DMEM/F12 containing 15% FBS was added for subsequent cell culture. Cell morphology was measured by microscopical measurement, and fibroblast identity was verified by detection of specific markers (α-SMA, FAP, and vimentin). All primary fibroblasts used in this study were less than three passages.

Co-culture assay

For co-culture experiments, CAFs or NFs were cultured with CRC cells using 0.4μm Transwell inserts (Corning, USA). Specifically, 6×10^4 CRC cells were seeded into the lower chamber, while 2×10^4 CAFs/NFs were plated in the upper chamber. After 48 hours of co-culture, CRC cells from the lower chamber were collected for following analyses.

Isolation and identification of exosomes

CAFs or NFs were cultured in serum-free DMEM/F12 medium for 48 hours to generate conditioned medium (CM). The CM was sequentially centrifuged at 4°C: first at 2,000 × g for 30 min (to remove dead cells), then at 10,000 × g for 30 min (to eliminate cell debris and large vesicles), and finally at 120,000 × g for 70 min (to pellet exosomes). The resulting pellet was washed with 1× PBS to reduce protein contamination, and filtered with 0.22-μm filters. Purified vesicles were characterized by Nanoparticle tracking analysis (NTA) using ZetaView PMX 110 ((Particle Metrix, Meerbusch, Germany) for size distribution,Transmission electron microscopy (FEI Tecnai G2 Spirit, Thermo Scientific, USA) for morphological validation, and Western blotting of exosomal markers.

Exosomes labeling and tracing

Isolated exosomes were fluorescently labeled with PKH67 (MCE, #257277-27-3), then added to the culture supernatant. After 24 hours of co-incubation with CRC cells, the nuclei were stained

with DAPI (Beyotime, #P0131), and exosome uptake was evaluated using confocal microscopy (Zeiss, Germany).

Fluorescence in situ hybridization (FISH)

Cellular localization of *circLDLR* was determined using FISH probe and kits (RiboBio, Guangzhou, China). Briefly, fixed and permeabilized cells were incubated with prehybridization buffer for 30 min at 37°C. Cy3-labeled *circLDLR* probe was mixed with pre-warmed hybridization buffer and incubated with cells overnight at 37°C in the dark. Post-hybridization washes were performed with wash buffer and PBS. Nuclei were stained with DAPI, and images were analyzed using confocal microscopy.

Proximity ligation assay (PLA)

The DuoLink® In Situ Red Starter Kit Mouse/Rabbit (DUO92101, Sigma-Aldrich) was used to detect interacting proteins. Briefly, CRC cells were fixed with 4% paraformaldehyde. The cells were washed with PBS and permeabilized using 0.1% Triton X-100 in PBS for 20 min. After blocking, the cells were incubated with primary antibodies at 4°C overnight. The subsequent step was incubating the pre-diluted anti-rabbit plus and anti-mouse minus probes at 37°C for 1h. Then, cells were incubated with 1× ligase for 30 min and 1× polymerase for 100 min at 37°C. Finally, coverslips were mounted on the slide with Duolink® In situ Mounting Medium with DAPI.

Western blot assay

Proteins extracted from cells or exosomes were separated by 10% SDS-PAGE and transferred onto 0.22μm PVDF membranes. Membranes were blocked with 5% fat-free milk for 1 hour at room temperature, followed by incubation with primary antibodies at 4°C overnight. After three washes with TBST, membranes were probed with species-matched secondary antibodies for 1 hour at room temperature. Following three additional TBST washes to remove nonspecific binding, target proteins were visualized using an Odyssey infrared imaging system (LI-COR Biosciences, USA). For lectin blot analysis, membranes were incubated with 20 μg/mL Vicia Villosa Lectin (VVL; Vector Labs, #B-1235-2) or Peanut Agglutinin (PNA; Vector Labs, #B-1075-5) in PBS for 30 min at room temperature. After wash with TPBS (PBS containing 0.05% Tween-20), membranes were incubated with ABC-HRP kit for 30 min at room temperature. Following additional TPBS washes, target glycoproteins were detected using enhanced chemiluminescence (ECL). The primary and secondary antibodies are listed in Table S4.

Immunoprecipitation (IP) and co-immunoprecipitation (CO-IP)

Total cellular proteins were extracted using Cell Lysis Buffer for Western and IP (Beyotime, #P0013). Firstly, 500 μL of antibody working solution or normal IgG working solution was added and incubated with A/G magnetic beads (MCE, #HY-K0202) for 1 hour at room temperature on a rotary mixer. Subsequently, 2 mg of total protein was incubated overnight at 4°C with protein A/G magnetic beads or protein-agarose bound VVL (Vectorlabs, #AL-1233). The magnetic beads or agarose beads were then washed three times with TBST, and the bound proteins were eluted using loading buffer for Western blot analysis.

RNA extraction and quantitative real-time PCR (qRT-PCR)

Total RNA from tissues and cells was isolated using TRIzol reagent (Invitrogen), while exosomal RNA was isolated with the miRNeasy Mini Kit (QIAGEN, #217004). Their complementary DNA

(cDNA) synthesis was performed using the PrimeScript RT reagent kit (TaKaRa, Shiga, Japan). qPCR was carried out with FastStart Universal SYBR Green Master Mix (ROX) (Roche, Switzerland) and primers (sequences in Table S5), using GAPDH as the endogenous control.

Immunohistochemistry (IHC) and hematoxylin–eosin (H&E) staining

Tissue samples were fixed in 4% paraformaldehyde (PFA), embedded, sectioned, and subsequently stained with various antibodies, including: Ki67(proteintech, #27309), GALNT14 (proteintech, #16939), 4HNE (Abmart, #PC6313) for IHC. For metastatic lesion analysis, liver and lung tissues were similarly fixed, embedded, sectioned, and stained with H&E. All images were captured using an Olympus microscope and processed using NDP.view2 imaging software.

CCK-8 assay

Cell viability was measured using the CCK-8 assay. Briefly, cells were seeded in 96-well plates at a density of 3×10^3 cells per well. At indicated time points, 10 μL of CCK-8 solution (SEVEN BIOTECH, #SC119) was added to each well, followed by incubation at 37°C with 5% CO_2 for 4 hours. Absorbance at 450 nm (OD_{450}) was measured using a microplate reader. Cell viability was assessed daily through OD_{450} measurements.

Colony formation assay

CRC cells were seeded at a density of 1500 cells/well into six-well plates for 2 weeks. Thereafter, the colonies were stained with 0.1% crystal violet and counted.

Migration and invasion assays

Falcon Transwell inserts (8-μm pore, Corning, USA) were used for migration assays. CRC cells were serum-starved for 24 hours, trypsinized, and resuspended in serum-free medium. Cells ($4×10^4$ in 200 μL serum-free medium) were seeded into the upper chamber of Transwell inserts. The lower chamber contained 700 μL medium with 10% FBS as chemoattractant. After 24 hours, non-migrated cells on the upper membrane surface were removed with cotton swabs. Migrated cells on the lower surface were fixed with 4% paraformaldehyde, stained with 0.1% crystal violet, and quantified. For the invasion assay, cells were inserted into the upper chamber of the Matrigel invasion chamber and subsequently treated as described for the transwell migration assay.

Lipid Peroxidation measurement

Lipid Peroxidation levels were assessed using the Lipid Peroxidation Assay Kit with BODIPY 581/591 C11 (Beyotime, #S0043S). Approximately 2 × 105 transfected cells/well were seeded in twelve-well plates overnight. Thereafter, cells were treated with 10μM erastin (MCE, #HY-15763) and 1μM liproxstatin-1 (Lip-1) (MCE, #HY-12726)for 48 h.BODIPY 581/591 C11 was added to cells in the dark for 20 min at 37°C, followed by washing with PBS. Thereafter, fluorescence imaging was performed using a ZEISS confocal microscope.

Iron assay

To quantify the intracellular levels of total iron and ferrous iron, an iron assay kit (APPLYGEN, #E1042) and a Ferrous Ion Assay Kit (Beyotime, #S1070S) were used according to the manufacturer's instructions.

Mitochondrial membrane potential (MMP) measurement

The MMP was measured using theJC-1 Mitochondrial Membrane Potential Assay Kit

(MCE, #HY-K0601), according to the manufacturer’s instructions.

Mitochondrial superoxide measurement

Mitochondrial superoxide levels were determined using MitoSOX Red mitochondrial superoxide indicator for live-cell imaging (MCE, #HY-D1055), according to the manufacturer’s instructions.

RNA pull-down assay

Biotinylated full-length RNA probes and fragmented *circLDLR* probes were synthesized by GenePharma Co., Ltd (Shanghai, China). Streptavidin Magnetic Beads (MCE, #HY-K0208) were pre-cleared by incubating with transfected cell lysates for 1 hour at 4°C to reduce non-specific binding. Biotinylated probes were then incubated with the beads for 60 minutes at room temperature to form probe-bead complexes. These complexes were subsequently incubated with transfected cell lysates overnight at 4°C. After magnetic separation, the beads were washed five times with buffer. RNA-protein complexes were eluted and analyzed by Western blotting. The probes used in RNA pull-down are provided in Table S5.

RNA immunoprecipitation (RIP)

RIP experiments were performed with a Magna RIP RNA-Binding Protein Immunoprecipitation Kit (Millipore, Billerica, MA, USA) according to the manufacturer's instructions. Co-precipitated RNA was detected by qRT-PCR. The primers used in RIP are provided in Table S5.

Mice xenograft model

All experimental procedures were approved by the Animal Ethics Committee of the First Affiliated Hospital of Harbin Medical University. Female 4–6 weeks old BALB/c nude mice were purchased from Huachuang Sino Pharma Tech Co., Ltd (Jiangsu, China) and maintained under SPF conditions in a controlled environment of 12 h of light and darkness, 50–70% humidity, 23 ± 2°C, and animals had free access to food and water. Mice were randomly grouped (n = 6 per group) after 1 week of acclimation, and the luciferase-labeled lentivirus was transfected into CAFs and selected with puromycin (2μg/ml) for 10 days. CRC (3×10^6) cells either alone or mixed with CAFs (Vector/sh-*circLDLR*/OE-*circLDLR*) at the ratio of 1:1 in 200 µl PBS were subcutaneously injected into nude mices. Tumors were measured every 5 d for 20 days, volume (mm^3) = length × width^2 × 0.52. The mice were sacrificed after 20 days and the tumors were excised and weighed.

For the lung metastasis models, Luciferase-labeled 1×10^6 HCT116 either alone or mixed with CAFs (Vector/sh-*circLDLR*) at the ratio of 1:1 in 100 µl PBS were injected into the tail veins of nude mice (n = 6/group). The mice were euthanized 6 weeks later, and the lungs were surgically dissected for further investigation.

To establish a liver metastasis models, Luciferase-labeled 2×10^6 RKO either alone or mixed with CAFs (Vector/OE-*circLDLR*) at the ratio of 1:1 in 200 µl PBS were injected into the spleens of nude mice. Briefly, nude mice (n = 6/group) were anesthetized, and a lateral incision was made to expose the spleen. Mixed cells were resuspended in 200 µL PBS and injected into the spleen using a microsyringe. The wounds were closed with 4 – 0 sutures. The mice were euthanized, and liver samples were excised for further analysis after 3 weeks.

Before the lung and liver metastasis model mice were euthanized, the IVIS Lumina imaging station (Caliper Life Sciences, Hopkinton, MA, USA) was used for bioluminescence imaging after intraperitoneal injection of 150 µl D-luciferin (15mg/ml) (Beyotime, #ST196). Lastly, the primary tumors of subcutaneous xenograft models, lungs and livers of metastasis models were obtained and fixed with 4% formalin. The level of ki67, GALNT14 and 4HNE were detected by IHC. The metastatic area in the lungs and livers were carefully examined by H&E staining.

Statistical analysis

All statistical analyses were performed using GraphPad Prism 8.0 (San Diego, CA, USA). Bioinformatics analysis was performed with R (version 4.2.1) (http://www.R-project.org/). The number of biological replicates for each experiment was indicated in the corresponding figure legend. All values represented the mean ± standard deviation (S.D) and were derived from a minimum of three independent biological replicates. *P* values are calculated by unpaired two-sided *t*-test. In figures, statistical comparisons between the control group and the experimental group are denoted as follows: ns no significance, $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$.

Declarations

Data availability

The data that support the findings of this study are available from the corresponding author on request.

COMPETING INTERESTS

The authors declare that they have no competing interests.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

This study was approved by the institutional ethics review board of the First Affiliated Hospital of Harbin Medicine University (no. IRB-AF/SC-04/02.2) and by the Animal Ethics Committee of the First Affiliated Hospital of Harbin Medicine University (no. 2023089). Participants gave informed consent to participate in the study before taking part.

AUTHOR CONTRIBUTIONS

H.Y.P supervised the project. H.Y.P and J.L.G conceived and designed the study. J.L.G, L.F.G and S.N performed experiments. J.L.G and W.X.Z contributed to data analysis and generated figures. H.Y.P and J.L.G wrote the manuscript. All authors reviewed and approved the manuscript for submission.

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Figures

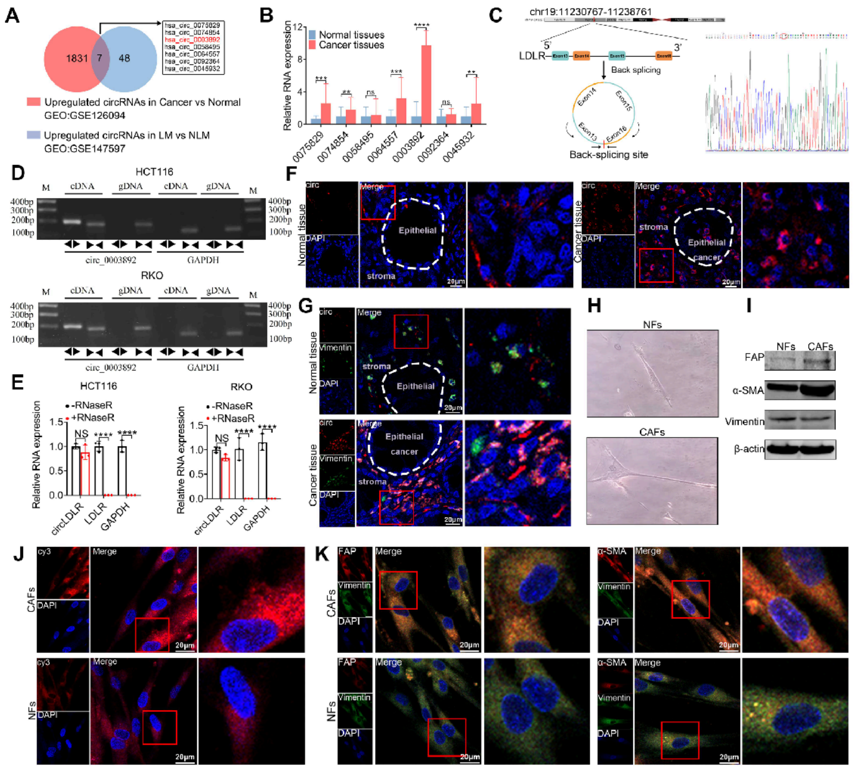

Fig.1 The expression and characteristics of *circLDLR*.

A. Two independent CRC datasets used to identify upregulated circRNAs in CRC from GEO database. **B**. The expression of 7 circRNAs was measured by RT-qPCR in 43 paired CRC tissues and adjacent normal tissues. **C-E.** Characteristics of circLDLR. *n*=3 biologically independent experiments. **F.** Analysis of RNA FISH for *circLDLR* in CRC tissue and Normal tissue. (Scale bar, 20 μm) **G.** RNA FISH and immunofluorescence co-localization showed the expression of *circLDLR* in CRC tissue. (Scale bar, 20 μm) **H.** Representative morphology of CAFs and NFs. **I.K.** Western blot and IF were used to detect the levels of α-SMA, FAP and Vimentin of CAFs and NFs. (Scale bar, 20 μm) **J.** FISH were used to compare the expression of *circLDLR* between CAFs and NFs. (Scale bar, 20 μm) $^*P<0.05$, $^{**}P<0.01$, $^{***}P<0.001$, $^{****}P<0.0001$. Data are presented as mean ± S.D. *P* values are calculated by unpaired two-sided *t*-test. *0075829* refers to *hsa_circ_0075829, 0003892* refers to *hsa_circ_0003892*，*0074854* refers to *hsa_circ_0074854, 0058495* refers to *hsa_circ_0058495, 0064557* refers to *hsa_circ_0064557, 0092364* refers to *hsa_circ_0092364, 0045932* refers to *hsa_circ_0045932.*

Figure 1

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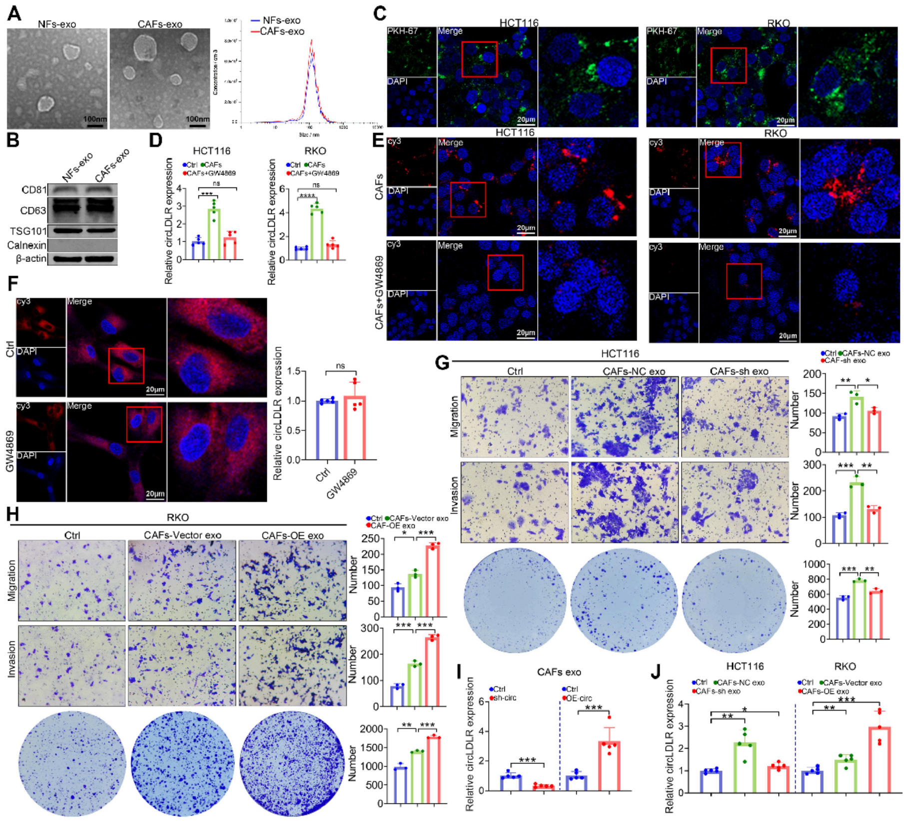

Fig. 2 CAFs promotes the progression of CRC cells through exosomes-transferred *circLDLR*.

A. Representative images of transmission electron micrograph and NTA of exosomes derived from CAFs and NFs, scale bar=100nm. **B**. Western blot analysis of exosome markers FAP, α-SMA, and Vimentin of exosomes derived from CAFs and NFs. **C.** CRC cells were incubated with exosomes labeled with PKH67, and then examined by confocal microscopy. (scale bar, 20 μm) **D, E.** qRT-PCR and RNA-FISH assays were used to detect the expression of *circLDLR*. (scale bar, 20 μm) **F.** The level of *circLDLR* in CAFs. (scale bar, 20 μm) **G, H.** The proliferation, migration and invasion abilities of CRC cells. *n*=3 biologically independent samples. **I, J.** Exosomes from overexpression or deletion of *circLDLR* in CAFs impacted the expression of *circLDLR* in CAF exosomes and cancer cells. ns no significance,**P*<0.05, ***P*<0.01,****P*<0.001. In **D**, **F**, **I** and **J**, *n*=5 biologically independent experiments. In **G**, **H**, *n*=3 biologically independent experiments. Data are presented as mean ± S.D. *P* values are calculated by unpaired two-sided *t*-test.

Figure 2

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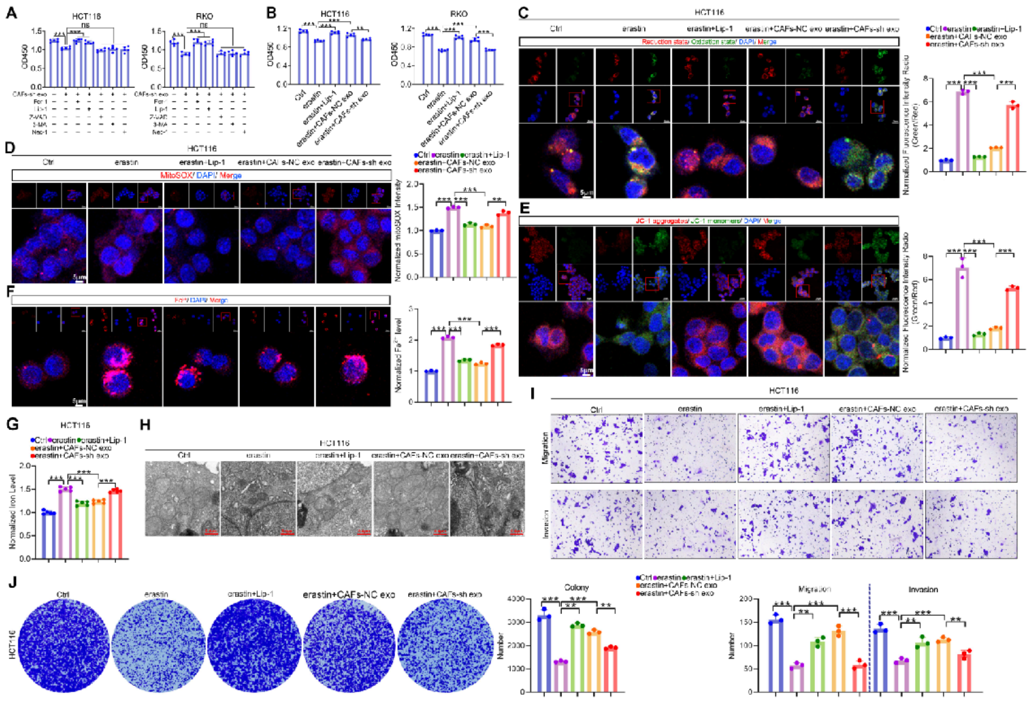

Fig. 3 CAFs-derived *circLDLR* facilitates the progression of CRC cells by suppressing ferroptosis.

A. Cell viability data from CCK-8 assay after treatment with various specific inhibitors of cell death. **B.** The proliferation ability of HCT116. **C.** Lipid peroxidation level in HCT116. (Scale bar, 5 μm) **D, E.** Relative levels of MitoSOX and MMP in HCT116. (Scale bar, 5 μm) **F, G.** Relative Fe2+ and Iron levels in HCT116. (Scale bar, 5 μm) **H.** TEM images of mitochondria. (Scale bar, 0.4 μm) **I, J.** Invasion, migration and colony formation assays. ns no significance,**P<0.01,***P<0.001. In **A**, **B** and **G**, n=5 biologically independent experiments. In **C-F**, **I** and **J**, n=3 biologically independent experiments. Data are presented as mean ± S.D. P values are calculated by unpaired two-sided t-test.

Figure 3

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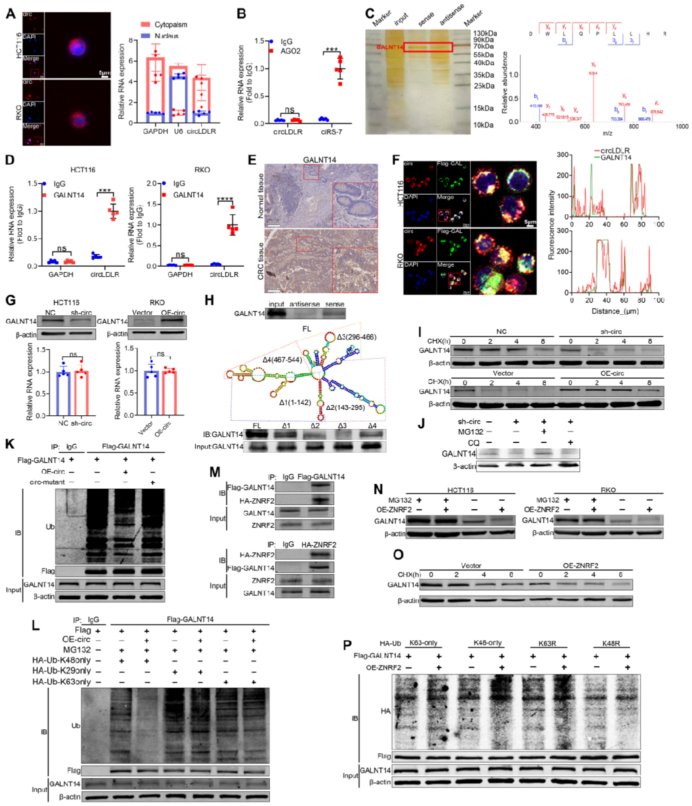

Fig. 4 ***CircLDLR*** **stabilizes GALNT14 via protecting it against ZNRF2-mediated ubiquitin/proteasome-mediated degradation.**

A. Cytoplasmic localization of *circLDLR* in CRC cells detected by FISH and qRT-PCR. (Scale bar, 5 μm) **B.** RIP assays for *circLDLR* and ciRS-7 enrichment. **C.** Circ-binding proteins identified by MS. **D.** RIP assays were used to confirmed the binding between *circLDLR* and GALNT14. **E**. The expression of GALNT14 in Normal tissue and CRC tissue. (Scale bar, 200 μm) **F.** The co-localized of *circLDLR* and GALNT14. (Scale bar, 5 μm) **G.** *CircLDLR* was positiveiy correlated with GALNT14 protein level but not mRNA level in CRC cells. **H.** Western blotting of RNA pull-down assays revealed the interaction between GALNT14 and *circLDLR* wildtype/mutants. **I.** CRC cells with *circLDLR* overexpression or deletion were treated with CHX (50μg/ml) for the indicated times, and then subjected to immunoblot assays of GALNT14 level. **J.** CRC cells were treated with CQ (10μM) or MG132 (10μM). Western blotting analysis of GALNT14 protein levels. **K.** Western blot assay showing the level of Ubiquitination on GALNT14 in 293T transduced with wild-type *circLDLR* or mutant *circLDLR* (mutant:GALNT14-binding sites). **L.** Western blotting assay of polyubiquitin chains of GALNT14 in *circLDLR* overexpressing 293T. **M, N.** Western blotting showed the relationship between ZNRF2 and GALNT14. **O.** ZNRF2 decreased the half-life of GALNT14 protein. **P.** ZNRF2 targets GALNT14 for K48-linked ubiquitination. HA-Ub-K48only: the cells were transfected with plasmids expressing HA-tagged ubiquitin with all lysines mutated except K48. HA-Ub-K63only: the cells were transfected with plasmids expressing HA-tagged ubiquitin with all lysines mutated except K63. HA-Ub-K29only: the cells were transfected with plasmids expressing HA-tagged ubiquitin with all lysines mutated except K29. HA-Ub-K48R: the cells were transfected with plasmids expressing HA-tagged ubiquitin with K48 mutated. HA-Ub-K63only: the cells were transfected with plasmids expressing HA-tagged ubiquitin with K63 mutated. ns no significance,***P<0.001,****P<0.0001. n=5 biologically independent experiments. Data are presented as mean ± S.D. P values are calculated by unpaired two-sided t-test.

Figure 4

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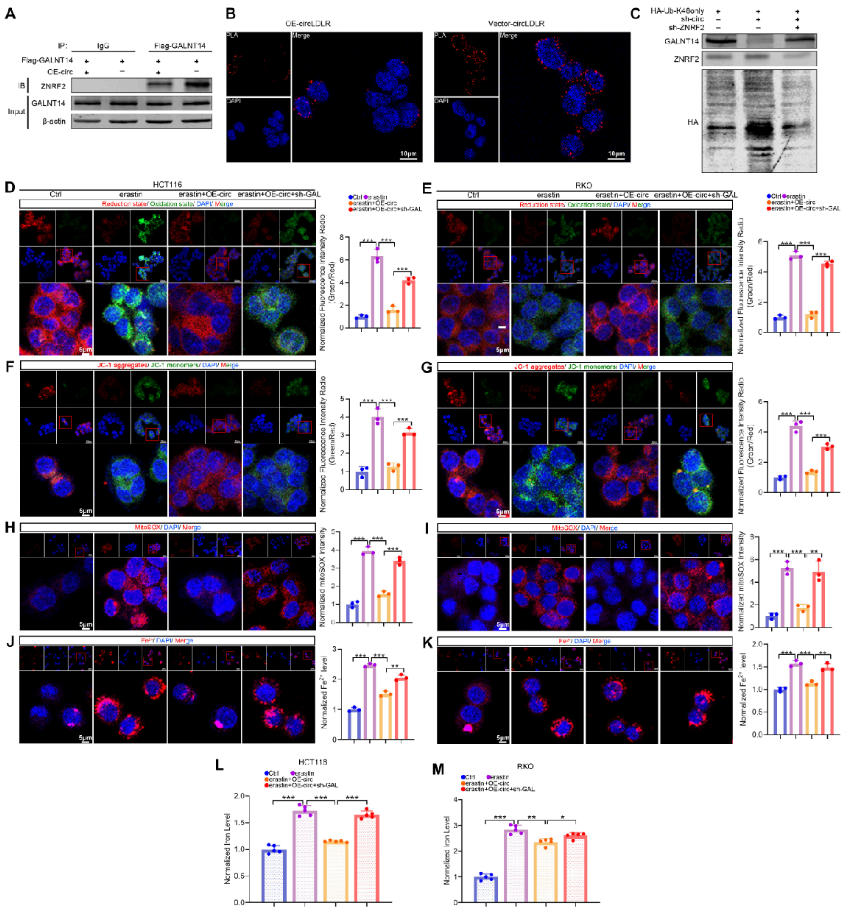

Fig. 5 *CircLDLR* confers ferroptosis resistance by upregulating GALNT14.

A. The overexpression effect of *circLDLR* on the interaction between GALNT14 and ZNRF2 determined by CO-IP assay. **B.** Tthe overexpression effect of *circLDLR* on the interaction between GALNT14 and ZNRF2 determined by PLA. **C.** The effect of ZNRF2 silencing on the GALNT14 caused by *circLDLR* deletion through the K48-linked ubiquitin proteasome pathway. **D, E.** Lipid peroxidation level in CRC cells. (Scale bar, 5 μm) **F-I.** Relative levels of MMP and MitoSOX in CRC cells. (Scale bar, 5 μm) **J, K.** Relative Fe^{2+} levels. (Scale bar, 5 μm) **L, M.** Relative Iron levels in CRC cells. Scale bar, 20 μm. n=3. **P*<0.05, ***P*<0.01, ****P*<0.001. In **D-K**, *n*=3 biologically independent experiments. In **L, M**, *n*=5 biologically independent experiments. Data are presented as mean ± S.D. *P* values are calculated by unpaired two-sided *t*-test.

Figure 5

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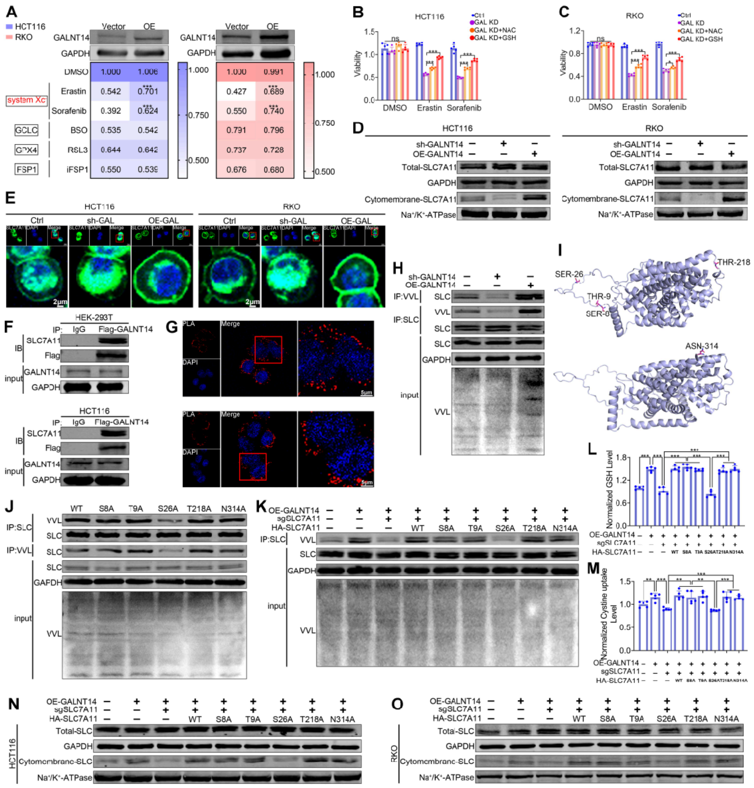

Fig. 6 GALNT14 impairs ferroptosis via O-GalNAcylation of SLC7A11 and its membrane localization.

A. Cell viability assay of GALNT14 overexpressed CRC cells added with different pathway ferroptosis inducers for 48 h. **B, C.** Cell survival assay of GALNT14 deleted CRC cells added with System Xc− ferroptosis inducers and NAC/GSH. **D, E.** Immunoblotting and immunofluorescence assays to detect the membrane localization of SLC7A11 of GALNT14 mutants in CRC cells. (scale bar, 2 μm) **F.** Relationship between SLC7A11 and GALNT14 determined by CO-IP assay. **G.** Relationship between SLC7A11 and GALNT14 determined by PLA. **H.** O-GalNAcylation level of SLC7A11 affected by GALNT14. **I.** The predicted O-glycosylation sites and N-glycosylation of SLC7A11. **J.** SLC7A11 with S8R/T9R/S26R/T218R/N314R mutation was expressed in indicated cells. Then the cells were performed with CO-IP and VVL lectin blot assays. **K.** Western blot assay of the O-GalNAcylation level of SLC7A11. **L, M.** The level of GSH and Cystine uptake in indicated cells. **N, O.** The membrane localization of SLC7A11 influenced by GALNT14. ns no significance,$^*P<0.05$, $^{***}P<0.001$. $n=5$ biologically independent experiments. Data are presented as mean ± S.D. P values are calculated by unpaired two-sided t-test.

Figure 6

See image above for figure legend

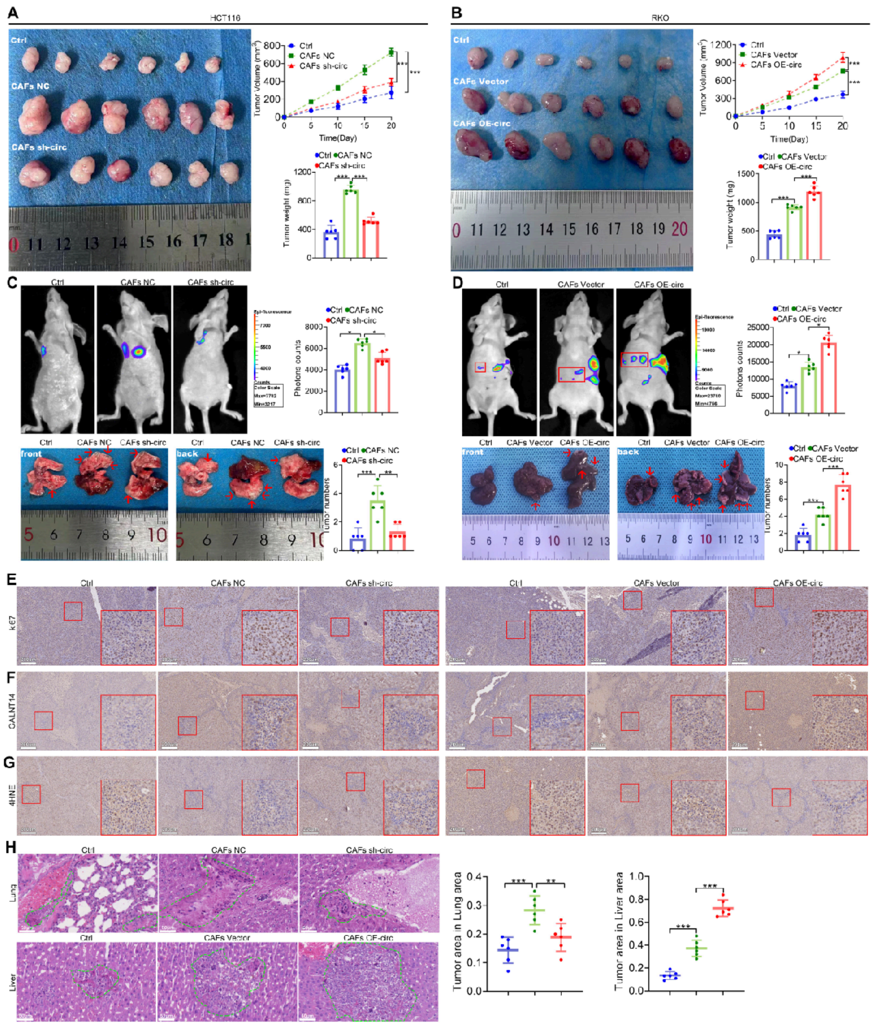

Fig. 7 *CircLDLR* in CAFs promote cancer growth and metastasis in vivo.

A, B. Nude mice were subcutaneously implanted with CRC cells co-injected with CAFs. **C.** Representative images of the lung tissues. **D.** Representative images of the liver tissues. **E-G**. IHC analysis of KI67, GALNT14 and 4HNE levels in subcutaneous tumors. (scale bar, 200 μm) **H.** H&E analysis in lung metastasis and liver metastasis tissues. (scale bar, 50 μm) $^{*}P<0.05$,$^{***}P<0.001$. $n=6$ biologically independent mice. Data are presented as mean ± S.D. P values are calculated by unpaired two-sided t-test.

Figure 7

See image above for figure legend

Supplementary Files

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